
1 Introduction

1.1 Background

With the expansion of the unmanned vehicle (land, air, surface and underwater) in many battlefield and disaster scenarios, there is a demand for methods of controlling these. Whilst autonomy is garnering a lot of attention in this way, there is a need for the expertise and skill that a human controller brings, whilst standing them off from the dangers of hazardous environments.

An area of great interest currently is in unmanned air vehicles, where development is being taken up in earnest by organisations from major engineering firms to undergraduate project groups. Another field which is developing rapidly, however, is that of marine autonomy. This is evidenced by the speech given by the First Sea Lord at the DSEI conference, where the need for a “fleet of crewed, uncrewed, and autonomous platforms” [1] was identified. Additionally in the merchant naval space, the uncrewed platforms are being recognised, with the Workboat Code Edition 3 adding Annexe 2 in 2025 to specifically target what it terms “Remotely Operated Unmanned Vessels (ROUVs)” in section 1.1.1 [2].

Companies such as Saab AB, Thales Group, and L3Harris [3][4][5] have been making progress towards realising the vision of a “hybrid fleet” whilst companies such as ACUA Ocean, Deep Ocean Engineering, and HD Hyundai [6][7][8] are investing in ROUVs for the commercial market.

1.2 Problem Statement

Autonomy in workboats is as such becoming increasingly mature and is being deployed to many different platforms. Until a level of assurance in the autonomous controller’s capabilities and reliability is reached (if such a level will ever be reached), human oversight of the vessel will be required in some form. Even in situations where autonomy is not the end goal for the vessel, there is a need for a method of removing the crew from the vessel in hazardous conditions. This can reduce the chance and/or severity of accidents (man overboard, injuries from machinery, loss of platform) whilst potentially improving quality of life for those crews (e.g. protection from the elements). Moreover, there is an opportunity for workboats to be constructed without a wheelhouse and on board controls, which could reduce the cost, complexity, and displacement of the platform. In all of these situations remote control of the vessel is desirable: to keep watch over autonomy or remove the crew from the vessel. As mentioned above, Workboat Code edition 3 Annexe 2 [2] prescribes requirements for an ROUV, but does not prescribe solutions to this issue.

1.3 System Narrative

In designing a remote control unit (RCU), the manner in which the operator issues commands and receives information is important. The RCU will have to provide situational awareness that is at least as good as a crew member would have aboard the vessel. It also must be able to provide the same functionality and level of control as those controls normally found aboard the platform. Additionally, it must be capable of connecting to the vessel in a reliable and secure way. Such a communication channel must be capable of providing an appropriate SIL, and must permit the transmission of commands, information, and alerts simultaneously. There are a number of approaches to this that are dependant on the setting in which the controller would be used.

There are also specific needs that an RCU will need to meet introduced by the remote control. The platform can only accept one controller at a time to avoid complexity in controller hierarchy. As such, the RCU will need to manage assuming control of the vessel, both from a “cold start” and from an exant controller. Additionally, the RCU will need to manage different methods of control behaviour (open water operation, confined spaces, and dynamic station keeping) whilst also being able to manage a degraded communication environment (such as in inclement weather).

Managing ROUV subsystems is also core functionality that must be met. Control of the propulsion and power systems is necessary for controlling movement and for enabling a “cold start” scenario. COLREGs 1972 mandates shapes, lights, and sounds that must be managed depending on the boat’s situation and movements, and must be controlled as well [9]. Additionally, the emergency system aboard the vessel must be controlled as well.

This includes both alerts to the operator and the emergency stop functionalities. To that end, the vessel must also comply with all relevant legislation, such as that pertaining to the ROUV and also to those regarding radio communications [2], [9], .

1.4 Aims and Objectives

This report aims to provide an initial architectural solution to a man portable RCU that can control an ROUV within line of sight in compliance with all relevant legislation, which can be then used to inform the design of a realised solution.

To do this, this report has the following objectives:

1. To identify scenarios, capabilities, and use cases for an RCU, whilst understanding the relationships between different use cases and the actors external to the system.
2. To define a set of user activities that the RCU would need to be capable of facilitating derived from the scenarios and use cases.
3. To derive a set of requirements from the activities, use cases, and relevant legislation, and to functionally decompose them to inform the structure of the RCU.
4. To identify the constituent parts of a solution, defining their properties and internal structures, allocating them to the functional requirements, and structuring them such that the RCU would be able to facilitate the user activities and scenarios.
5. To supplement the behavioural model of the architected solution with respect to the structural model, to further understand the interaction between the actors and subsystems.

2 Technical Results

2.1 Functional Requirements Definition

2.2 Functional Decomposition, Elaboration, and Allocation

2.3 System Structure Definition

2.4 System Interface Behaviour

2.5 Alternative Behaviours

2.6 Architecture Analysis

3 Conclusion

3.1 System Specification

3.2 Achievement of Aims and Objectives

3.3 Highlights and Recommendations

References

- [1] G. Jenkins, *First sea lord general sir gwyn jenkins speech at dsei 2025*, Sep. 2025. [Online]. Available: <https://www.gov.uk/government/speeches/first-sea-lord-general-sir-gwyn-jenkins-speech-at-dsei-2025>.
- [2] Ship Standards, “Workboat code edition 3,” Maritime and Coastguard Agency, Tech. Rep., Jan. 2025. [Online]. Available: www.gov.uk/government/organisations/maritime-and-coastguard-agency.
- [3] Saab AB. “Naval autonomy | naval | saab.”[Online]. Available: www.saab.com/products/naval/naval-autonomy.
- [4] Thales Group. “Naval drone warfare | thales group.”[Online]. Available: <https://www.thalesgroup.com/en/solutions-catalogue/defence/naval/naval-drone-warfare>.
- [5] L3Harris. “Autonomous systems | l3harris® fast. forward.”[Online]. Available: <https://www.l3harris.com/all-capabilities/autonomous-systems>.
- [6] ACUA Ocean. “Home | acua ocean | uncrewed surface vessel | usv | plymouth, uk.”[Online]. Available: www.ocean.tech.
- [7] Deep Ocean. “Underwater rovs, usvs - underwater drones - deep ocean engineering.”[Online]. Available: www.deepocean.com.
- [8] HD Hyundai Co. Ltd. “Avikus | hd hyundai.”[Online]. Available: www.hd.com/en/business/shipbuilding/avikus/contents.
- [9] International Maritime Organization, “1972 convention on the international regulations for preventing collisions at sea,” International Maritime Organization, Tech. Rep., Oct. 1972. [Online]. Available: www.cil.nus.edu.sg.